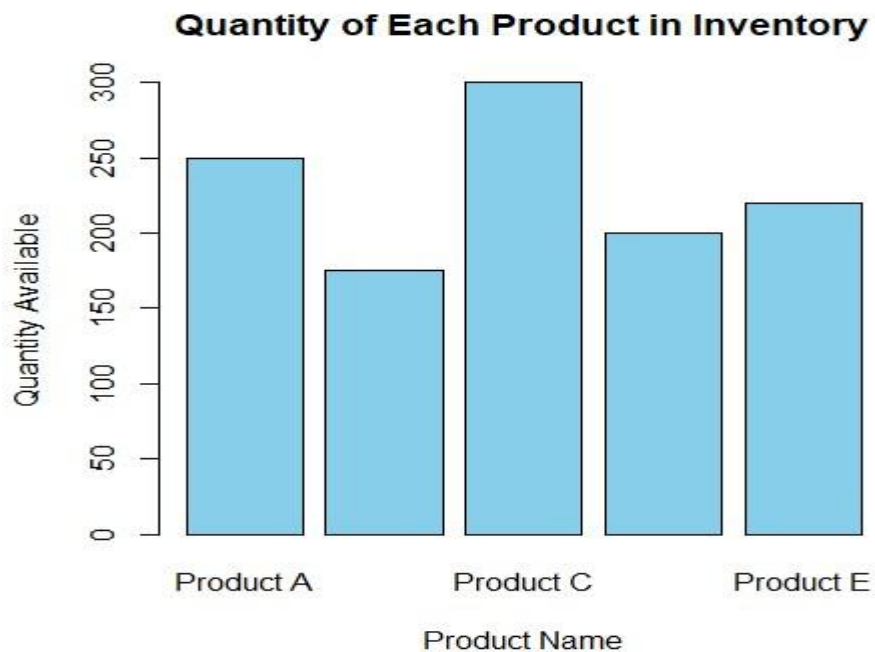


SET 20 - Product Inventory Management

20 A. Bar Chart

Program:

```
inventory <- data.frame(  
  Product_ID = c(1,2,3,4,5),  
  Product_Name = c("Product A","Product B","Product C","Product  
D","Product E"), Quantity = c(250,175,300,200,220)) print(inventory)  
barplot( inventory$Quantity, names.arg = inventory$Product_Name,  
col = "skyblue", main = "Quantity of Each Product in Inventory", xlab =  
"Product Name", ylab = "Quantity Available")
```



20 B. Stacked Bar Chart – Quantity by Product Category

Program:

```
# Add categories inventory$Category <-  
c("Electronics","Electronics",  
  "Furniture","Furniture",  
  "Accessories")  
# Create table stack_data <- xtabs(Quantity ~ Category +  
Product_Name, data = inventory)
```

```
# Stacked bar chart barplot( stack_data,
col =
c("red","blue","green","orange","purple
"), main = "Quantity by Product
Category", xlab = "Category", ylab =
"Quantity Available", legend =
rownames(stack_data))
```

Output:



20 D. Scatter Plot – Product Price vs Quantity Available

Program:

```
inventory$Price <- c(1200,1500,1000,1800,1400)
plot( inventory$Price,
inventory$Quantity, pch = 19, col =
"blue", main = "Product Price vs
Quantity Available", xlab = "Product
Price", ylab = "Quantity Available")
text( inventory$Price,
inventory$Quantity,
labels =
inventory$Product_Name,
```

pos = 3, cex = 0.8)

Output:

